

DIGITAL LOGIC DESIGN

ASSIGNMENT #03

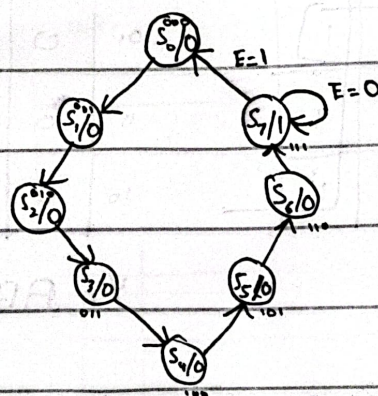
BEE-4A

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22L-7638

Q. No.1

State Diagram:-



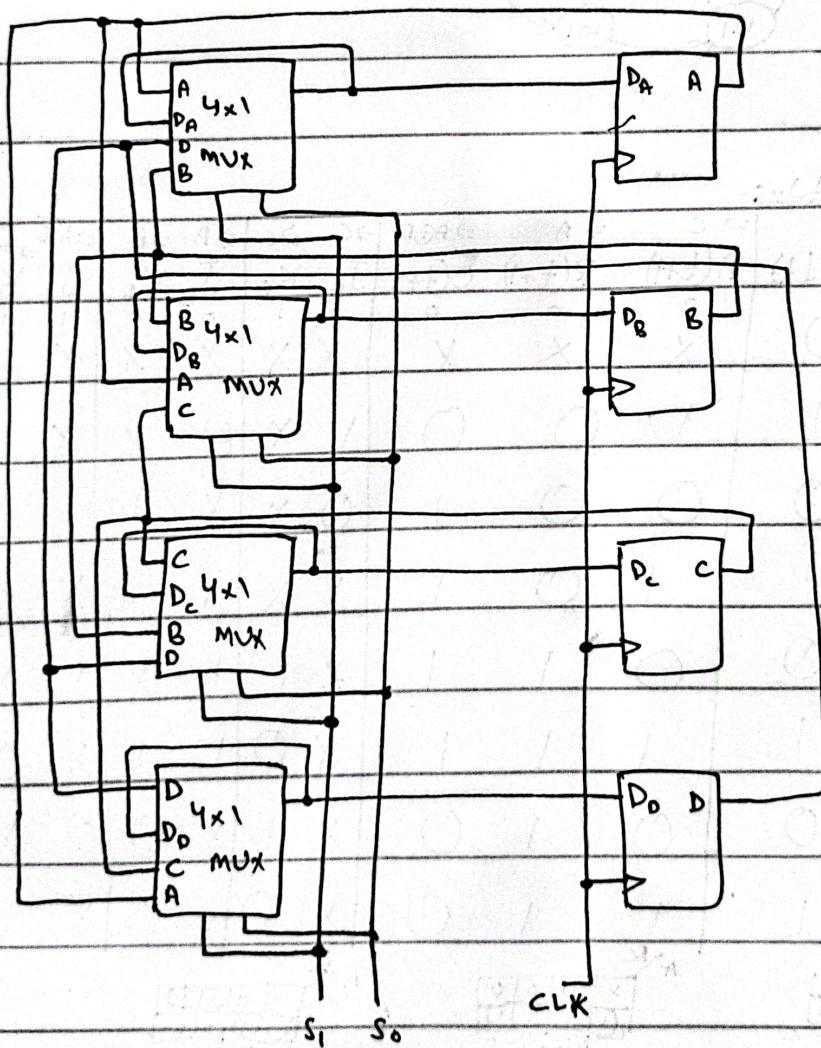
⇒ Moore Model

State Table:-

State Table:									
Present States			Next States			Output y			
A	B	C	E=0				E=1		
			D _A	D _B	D _C		D _A	D _B	D _C
0	0	0	0	0	1	0	0	1	0
0	0	1	0	1	0	0	1	0	0
0	1	0	0	1	1	0	1	1	0
0	1	1	1	0	0	1	0	0	0
1	0	0	1	0	1	1	0	1	0
1	0	1	1	1	0	1	1	0	0
1	1	0	1	1	1	1	1	1	0
1	1	1	1	1	1	0	0	0	1

Q. No. 2

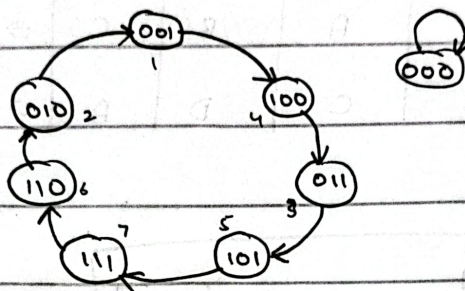
S_1	S_0	D_A	D_B	D_C	D_D	
0	0	A	B	C	D	⇒ No change
0	1	D_A	D_B	D_C	D_D	⇒ Parallel load
1	0	D	A	B	C	⇒ Shift Right
1	1	B	C	D	A	⇒ Shift Left



Q. No. 3

Sequence: 1, 4, 3, 5, 7, 6, 2, 1

State Diagram:-



State Table:-

A(t)	B(t)	C(t)	$\Rightarrow C$	$\Rightarrow A$	$\Rightarrow A \oplus B$	$\Rightarrow C$	$\Rightarrow \bar{C}$	$\Rightarrow A$	$\Rightarrow \bar{A}$	$\Rightarrow A \oplus B$	$\Rightarrow A \oplus B$
			A(t+1)	B(t+1)	C(t+1)	J _A	K _A	J _B	K _B	J _C	K _C
0	0	0	X	X	X	X	X	X	X	X	X
0	0	1	1	0	0	1	X	0	X	X	1
0	1	0	0	0	1	0	X	X	1	1	X
0	1	1	1	0	1	1	X	X	1	X	0
1	0	0	0	1	1	X	1	1	X	1	X
1	0	1	1	1	1	X	0	1	X	X	0
1	1	0	0	1	0	X	1	X	0	0	X
1	1	1	1	1	0	X	0	X	0	X	1

$A \oplus B$

X	1	1	0
X	1	1	0

$A(t+1) = C$

$A \oplus B$

X	0	0	0
1	1	1	0

$B(t+1) = A$

$A \oplus B$

X	0	0	0
1	1	1	0

$C(t+1) = A\bar{B} + \bar{A}B = A \oplus B$

As outputs are dependent on present states,

hence the above model is Moore.

		J_A			
A	BC	00	01	11	10
	0	X	1	1	0
	1	X	X	X	X

$$J_A = C$$

		K_A			
A	BC	00	01	11	10
	0	X	X	X	X
	1	1	0	0	1

$$K_A = \bar{C}$$

		J_B			
A	BC	00	01	11	10
	0	X	0	X	X
	1	1	1	X	X

$$J_B = A$$

		K_B			
A	BC	00	01	11	10
	0	X	X	1	1
	1	X	X	0	0

$$K_B = \bar{A}$$

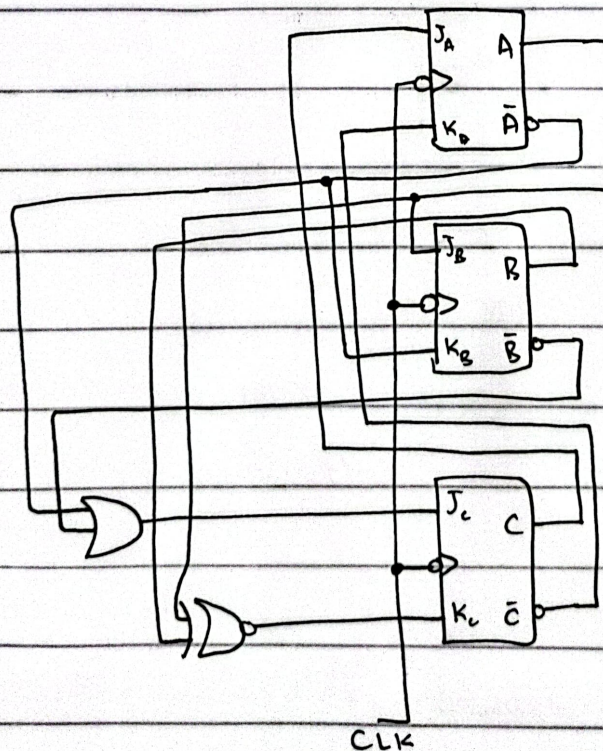
		J_C			
A	BC	00	01	11	10
	0	X	X	X	1
	1	1	X	X	0

$$J_C = \bar{B} + \bar{A}$$

		K_C			
A	BC	00	01	11	10
	0	X	1	0	X
	1	X	0	1	X

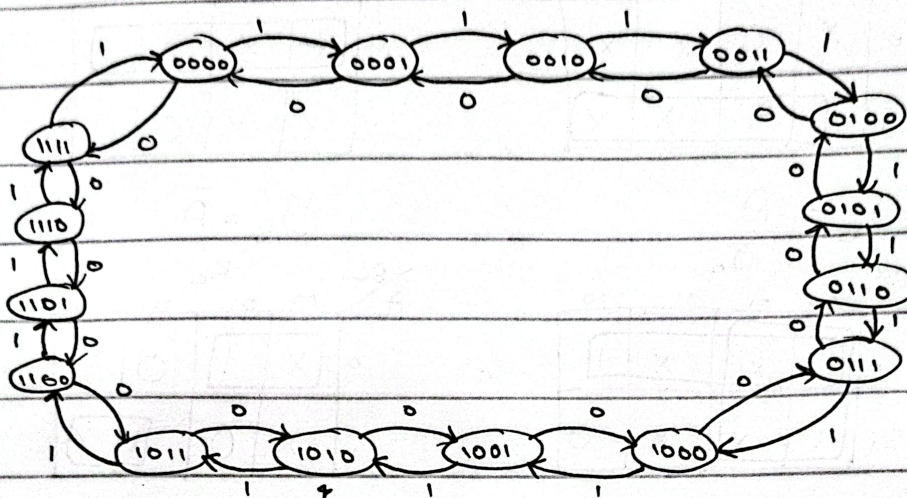
$$K_C = \bar{A}\bar{B} + AB$$

$$= A \oplus B$$



Q. No. 4

State Diagram :-



C	A	B	C	D	D _A	D _B	D _C	D _D
0	0	0	0	0	1	1	1	1
0	0	0	0	1	0	0	0	0
0	0	0	1	0	0	0	0	0
0	0	0	1	1	0	0	0	0
0	0	1	0	0	0	1	1	1
0	0	1	0	1	0	1	1	1
0	0	1	1	0	0	1	1	1
0	0	1	1	1	0	1	1	1
0	1	0	0	0	1	0	0	0
0	1	0	0	1	1	0	0	0
0	1	0	1	0	1	0	0	0
0	1	0	1	1	1	0	0	0
0	1	1	0	0	1	1	1	1
0	1	1	0	1	1	1	1	1
0	1	1	1	0	1	1	1	1
0	1	1	1	1	1	1	1	1
1	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0
1	0	0	1	0	0	0	0	0
1	0	0	1	1	0	0	0	0
1	0	1	0	0	0	1	1	1
1	0	1	0	1	0	1	1	1
1	0	1	1	0	0	1	1	1
1	0	1	1	1	0	1	1	1
1	1	0	0	0	1	0	0	0
1	1	0	0	1	1	0	0	0
1	1	0	1	0	1	0	0	0
1	1	0	1	1	1	0	0	0
1	1	1	0	0	1	1	1	1
1	1	1	0	1	1	1	1	1
1	1	1	1	0	1	1	1	1
1	1	1	1	1	1	1	1	1

③

C/D	0	1
0	1	0
1	0	0

 $D_B = \bar{C}\bar{D}$

④

C/D	0	1
0	0	1
1	1	1

 $D_B = C + D$

⑤

C/D	0	1
0	0	0
1	0	1

 $D_B = CD$

⑥

C/D	0	1
0	1	1
1	1	0

 $D_B = \bar{C} + \bar{D}$

⇒ Using 8x1 MUXs

From table, it is clear

$$\Rightarrow D_D = \bar{D}$$

$$\Rightarrow D_C = C \oplus D \text{ for first 4 inputs}$$

$$D_C = C \oplus D \text{ for last 4 inputs}$$

①

C/D	0	1
0	1	0
1	0	1

 $D_C = \bar{C}\bar{D} + CD = C \oplus D$

②

C/D	0	1
0	0	1
1	1	0

 $D_C = C\bar{D} + \bar{C}D = C \oplus D$

